32X1 MUX using 8X1 MUX and 4X1 MUX Verilog HDL

```
////////// 4x1 MUX /////////

module M4x1 (

  input [3 : 0]d,
  input [1 : 0]sel,
  output out

);

  assign out = sel[1] ? (sel[1] ? d[3] : d[2]) : (sel[0] ? d[0] : d[2]);

endmodule


///////////////// 8x1 MUX /////////////////

module M8x1 (

  input [7 : 0]y,
  input [2 : 0]sel,
  output out

);

  assign out = sel[2] ? (sel[1] ? (sel[0] ? y[7] :y[6]) :
                                  (sel[0] ? y[5] : y[4])
                        ) :
    (sel [1] ?
     (sel[0] ? y[3] : y[2]):
     (sel[0] ? y[1] : y[0]));

endmodule



////////// 32x1 MUX  through 8x1 MUX and 8x1MUX ///////////////

module MUX32x1 ( i, sel, out);

  input [31 : 0]i;
```

```
  input [4 : 0]sel;
  wire [3 : 0]w;
  output out;


  M8x1 A(i[31 : 24], sel[4 : 2], w[3]);
  M8x1 B(i[23 : 16], sel[4 : 2], w[2]);
  M8x1 C(i[15 : 8], sel[4 : 2], w[1]);
  M8x1 D(i[7 : 0], sel[4 : 2], w[0]);

  M4x1 E(w[3:0], sel[1:0], out);


endmodule

 /////////// testbench for 32x1 Mux ///////////

module mux32x1_tb;

  reg [31 : 0]i;
  reg [4 : 0]sel;
  wire out;

  integer u;

  MUX32x1 dut (i, sel, out);

  initial
    begin
      $dumpfile("MUX32x1.vcd");
      $dumpvars(0,mux32x1_tb);

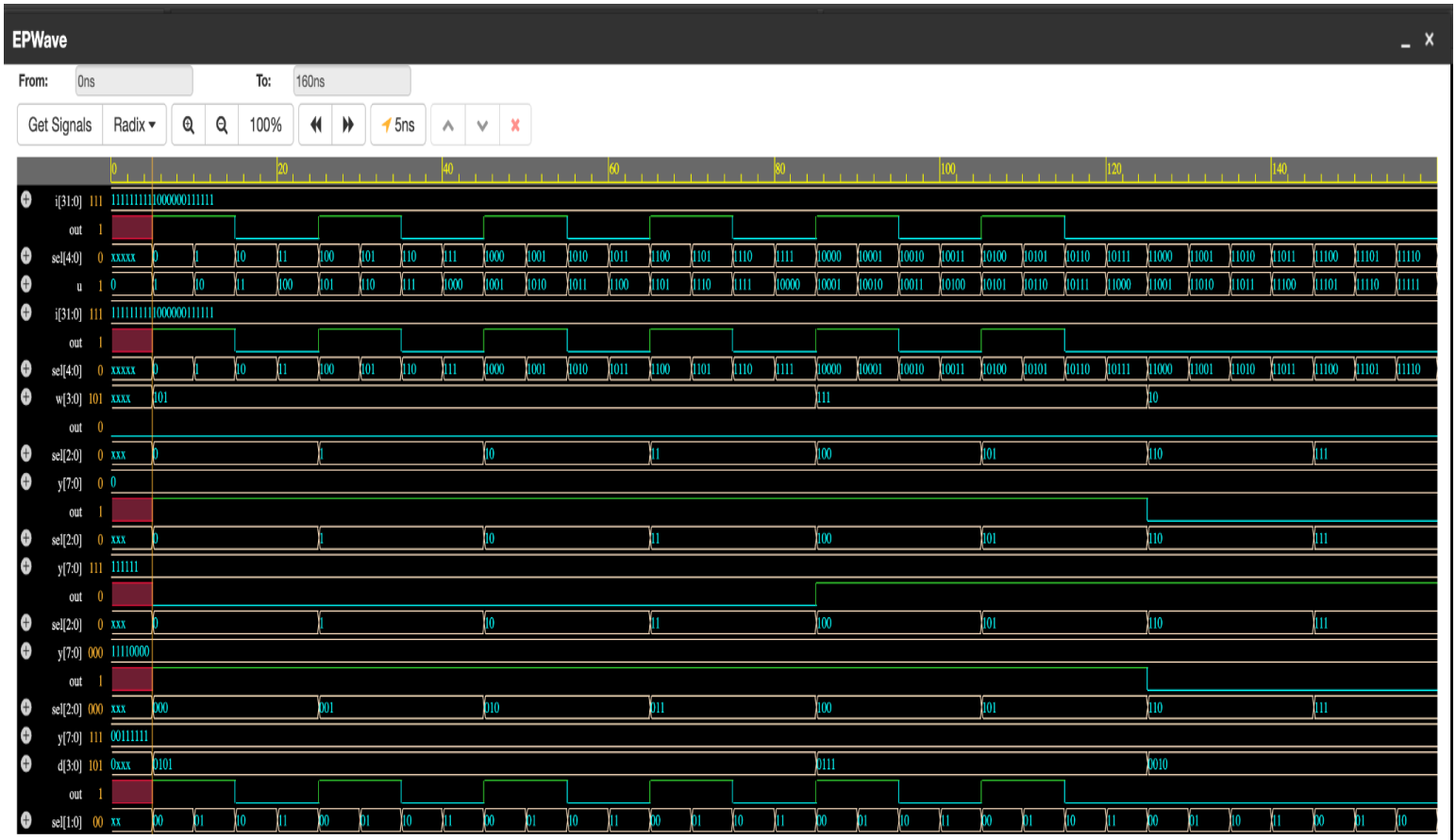
      monitor(time," | sel = %b | out = %b ", sel, out);

  begin

    i =  32'b00000000001111111111000000111111;

    for(u = 0; u < 32; u++)
      begin
        #5;
        sel = u;
      end
  End
```

```
    end
endmodule
```



Note: To revert to EPWave opening in a new browser window, set that option on your profile page.